Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Black Sea Production Centre
BLKSEA_ANALYSISFORECAST_BGC_007_010

Issue: 6.0

Contributors: L. Vandenbulcke, L. Macé, M. Grégoire, E. Peneva, R. Lecci

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	April 2019	All	First version for product 007_010 released at V201907	L. Vandenbulcke, A. Capet, M. Grégoire	
1.1	May 2019	All	Final version for qualification of 007_010 product	L. Vandenbulcke, A. Capet, M. Grégoire	E. Peneva
2.0	April 2020	All	New version V202007 of the product	L. Vandenbulcke, A. Capet, M. Grégoire	E. Peneva
2.1	April 2021		New dataset with optical variables	L. Vandenbulcke, A. Capet, M. Grégoire	E. Peneva
3.0	August 2022		New version V202211 of the product	L. Vandenbulcke, A. Capet, L. Mace, C. Meulders, A. Mouchet, M. Grégoire	E. Peneva
4.0	November 2023	All	New version V202311 of the product	L. Vandenbulcke, L. Macé, C. Meulders, A. Mouchet, M. Grégoire	E. Peneva
5.0	November 2024	All	New version V202411 of the product	L. Vandenbulcke, L. Macé, M. Grégoire	E. Peneva
6.0	May 2025	All	New version V202511 of the product	L. Vandenbulcke, M. Grégoire	E. Peneva, R. Lecci

ACRONYM TABLE

Acronym	Description
TAC	Copernicus Marine Service Thematic Assembly Centre
NRT	Near Real Time
REP	Reprocessing
MY	Multi-year
SST	Sea Surface Temperature
Argo	International profiling float network (www.argo.net)
BGC-ARGO	Biogeochemical Argo
EAN	Estimated Accuracy Number
EOF	Extended Orthogonal Function
MLD	Mixed Layer Depth
CIL	Cold Intermediate Layer
CCC	Cold Content in the CIL
DCM	Deep Chlorophyll maximum
DOX	Dissolved oxygen
CHL	Chlorophyll
NWS	North-Western Shelf (of the Black Sea)

KEY DEFINITIONS

MLD	Various criteria exist to define the MLD. In the Black Sea, the MLD is usually defined as the depth at which density is 0.125 $kg.m^{-3}$ larger than at the reference depth (usually 3m, but 10m in the current validation exercice)
CIL	In the Black Sea, a layer of water that is colder than the water above and below. It is considered to have less than 8.35°C.

RELEVANT DOCUMENT LIST

N/A

TABLE OF CONTENTS

Contents

ACRONYM TABLE 3
KEY DEFINITIONS 3
RELEVANT DOCUMENT LIST 3
I EXECUTIVE SUMMARY 5
I.1 Products covered by this document 5
I.2 Summary of the results 5
I.3 Estimated Accuracy Numbers 6
II PRODUCTION SYSTEM DESCRIPTION 7
II.1 Production centre details 7
II.2 Description of the production system 7
II.3 Upstream data and boundary condition of the NEMO-BAMHBI model 9
III VALIDATION FRAMEWORK 10
IV VALIDATION RESULTS 13
IV.1 Physical variables 13
IV.1.1 Mixed layer depth (MLD) 13
IV.1.2 Cold intermediate layer 14
IV.2 Oxygen 15
IV.3 Chlorophyll 19
IV.3.1 Satellite data 19
IV.3.2 In situ data 21
IV.4 Carbonate sub-system 21
IV.4.1 Oxygen and Nitrate 22
IV.4.2 Dissolved Inorganic Carbon 23
IV.4.3 pC02 25
IV.4.4 Air-Sea CO2 Flux 25
IV.5 Optics 27
IV.5.1 Photosynthetically Active Radiation (PAR) 27
IV.5.2 Optical attenuation coefficient (Kd) 29
IV.5.3 Reflectances 33
V SYSTEM'S NOTICEABLE EVENTS, OUTGAES OR CHANGES 35
VI QUALITY CHANGES SINCE PREVIOUS VERSION 37
VII References 38

I EXECUTIVE SUMMARY

The V202511 release of the Black Sea biogeochemical analysis-forecast uses the physical model NEMO v4.2, which is online-coupled with the BiogeochemicAl Model for Benthic and Hypoxic Influenced areas (BAMHBI). Compared to V2024, the river freshwater flux for all rivers except the Danube is now obtained from the EFAS platform[1], version 5. Similarly, the climatologic atmospheric deposition of nitrogen (nitrates, ammonium) has been replaced by the real-time product from the Copernicus Atmospheric Service (CAMS)[2].

The complete list of changes is given in Section VI.

I.1 Products covered by this document

BLK_ANALYSISFORECAST_BGC_007_010 is the biogeochemical component of the Black Sea analysis and forecast system implemented by the Black Sea Marine Forecasting Centre (BLK-MFC). It covers the Black Sea (approx. 41 to 47 °N, 28 to 42 °E) without the Sea of Azov (see Fig. 1). The online-coupled model is run with a horizontal resolution of approximately 2.5 km, 59 vertical levels, and generates the following six datasets:

- NUTR: Phosphorus, nitrate
- PFTC: Chlorophyll and phytoplankton biomass;
- BIOL: Dissolved oxygen (O_2) concentrations and net primary production (NPP); the BIOL dataset also includes the bottom dissolved oxygen concentration on the North-Western Black Sea shelf (regions 4 and 5 in Fig. 1), saved as a separate 2D variable.
- CARB: pH, dissolved inorganic carbon (DIC), total alkalinity (TA)
- CO2F: Surface partial CO_2 pressure (spCO2), surface CO_2 flux
- OPT: photosynthetically active radiation (PAR) and attenuation coefficient (Kd)

These datasets are provided as daily, monthly and yearly means, and for some variables (surface pCO_2, surface CO_2 flux), also as hourly means.

The companion multi-year (MY) product BLKSEA_MULTIYEAR_BGC_007_005, covering 1992-2024, is obtained by a similar forecasting system as the Near Real-Time (NRT) product validated in this document.

I.2 Summary of the results

The Black Sea biogeochemical NRT product has been validated using satellite chlorophyll[3] time series and *in situ* BGC-ARGO [4](chlorophyll, oxygen, PAR, Kd) data collected during the period 2023-2024. The other variables (6 dataset mentioned above) cannot be validated in NRT due to the absence of NRT data. They are validated in the context of the corresponding multi-year product (see BLK_MULTIYEAR_BGC_007_005) and additionally, their consistency with historical data is qualitatively

[1] https://european-flood.emergency.copernicus.eu/
[2] https://atmosphere.copernicus.eu/data
[3] https://doi.org/10.48670/moi-00301
[4] https://doi.org/10.48670/moi-00033

assessed for inorganic nutrients, DIC, pH, pCO_2, and TA. The validation metrics are described in section III, table 2.

Physics: The model is able to simulate the mixed layer depth (MLD), which is a key parameter resulting from physical processes that control the distribution of biogeochemical variables, water column stability, and ventilation mechanisms.

Chlorophyll: Since V202007 (see Change record table), surface chlorophyll is assimilated in the model. In the present version, V202511, the bias is low: -0.018 to 0.35 $mg.m^{-3}$ depending on the region (except near the Danube and Dnestr river mouths where bias is 0.44 to 1.10 $mg.m^{-3}$).
The depth of the deep chlorophyll maximum (DCM) is also well simulated.
Oxygen: The comparison of model and BGC-ARGO data reveals a bias of 1.6 $mmol/m^3$ and a Pearson correlation coefficient of 0.97. The Nash-Sutcliffe efficiency is a metric that indicates how large the difference between model and observation is, relative to the variability inside the observations. For oxygen, it is 0.94 ("excellent").
Carbonate system: DIC, TA, pH, pCO_2 and air-sea CO_2 flux have been compared to climatologies and historical values, and are all within the range of accepted values.
Optics: The V202411 includes a radiative transfer model. Incoming radiation is also obtained from an atmospheric spectral model. Photosynthetically Active Radiation (PAR) and Kd are compared with in situ (BGC-ARGO) and satellite data. The Estimated Accuracy Numbers (EANs) in Table 1 reveal that the normalized PAR is extremely well represented, and that the comparison between the product estimations and the BGC-ARGO measurements for Kd is very satisfactory.

I.3 Estimated Accuracy Numbers

This section includes a selection of the Estimated Accuracy Numbers for the product, including name of the variables with their units, the metric name as well as the values for each metric (Table 1). The metrics (bias, percentage bias, root mean square difference (RMSD), Pearson correlation and Nash-Sutcliffe efficiency) are described in section III.

Table 1: *Selected estimated accuracy numbers (EANs). We use different statistics to assess different aspects of model performance (e.g., variability, accuracy).*

Variable	Metric	Unit	Value
Oxygen (in situ)	Bias RMSD Nash-Sutcliffe Pearson correlation	mmol m^{-3} mmol m^{-3} - -	1.6 15.9 0.94 0.97
Oxygen profile maximum (in situ)	Bias	mmol m^{-3}	16.3
Chlorophyll (satellite) for deep sea for coastal areas	Bias	mg/m^3	 -0.019 to 0.019 -0.07 to 1.10
DCM depth (in situ)	Bias RMS	m m	23.89 27.29
Normalized PAR	Percentage bias STD ratio Nash-Sutcliffe Pearson correlation	% - - -	-12.99 1.16 0.79 0.89

II PRODUCTION SYSTEM DESCRIPTION

II.1 Production centre details

a) Production centre name: MARINES, Belgium

b) Production system name: BLK-MFC-Biogeochemistry (CMEMS name: BLKSEA_ANALYSISFORECAST_BGC_007_010)

c) Production centre description for the version covered by this document

The biogeochemical hindcasts and forecasts for the Black Sea are produced by the MARINES Production Unit using the NEMO v4.2 circulation model, online coupled with the BAMHBI biogeochemical model and the "Ocean Assimilation Kit" (OAK) data assimilation framework. The nominal production runs at the Cenaero (Belgium) Tier-1 supercomputing centre called Lucia. When Lucia is unavailable, backup production runs on servers rented from the Amazon cloud.

II.2 Description of the production system

Description of the Black Sea BIO operational system.

The Black Sea biogeochemical model (BLK-Biogeochemistry) is the Biogeochemical Model for Hypoxic and Benthic Influenced areas (BAMHBI, Gregoire et al., 2008; Grégoire and Soetaert, 2010; Capet et al., 2016b). It describes the food web from bacteria to gelatinous carnivores using 24 state variables including three phytoplankton groups (diatoms, small phototrophic flagellates and dinoflagellates), two zooplankton groups (micro- and mesozooplankton), two gelatinous zooplankton groups (omnivorous and carnivorous forms), and an explicit representation of the bacterial loop (bacteria, labile and semi-labile dissolved organic matter, particulate organic matter). The model simulates oxygen, nitrogen, silicate and carbon cycling. In addition, an innovation of this model is its explicit representation of processes in the anoxic layer.

Biogeochemical processes under anaerobic conditions are represented using an approach similar to the modelling of diagenetic processes in sediments, where all reduced substances are lumped into one state variable. This coupling ensures that processes in the upper oxygenated layer interact fullywith anaerobic processes in the deep waters, which is absolutely necessary for long-term simulations. Key low-oxygen processes such as denitrification, anaerobic ammonium oxidation (ANAMMOX), and reduced decomposition efficiency are explicitly represented (Gregoire et al., 2008). Moreover, the model includes a representation of diagenetic processes (Capet et al., 2016b) using an efficient and economic representation proposed by Soetaert et al. (2000). The benthic module represents the impact of sediment processes on key biogeochemical cycles. It includes sediment oxygen consumption, which drives summer hypoxia; degradation of organic matter, processing ~30 % of shelf primary production; and nitrate removal by benthic denitrification, filtering ~50 % of nitrogen from northwestern shelf rivers (mainly the Danube) and modulating deep-basin primary production. In addition to a representation of diagenesis, the biogeochemical model simulates sediment transport by waves. This is an important feature for sustaining primary production in the deep basin.

Since V201907, BAMHBI has been extended with a module describing the carbonate dynamics. Following Soetaert et al. (2007), the model state variables include DIC and the "excess negative charge", from which total alkalinity (TA) is computed, taking in account the contribution of sulphide, pH, the speciation of DIC ($[HCO_3]^-$, $[CO_3]^{2-}$, $[CO_2]$), and the air-sea CO_2 flux.

Since V202007, the forecasting system includes data assimilation of daily level-3 satellite observations of surface chlorophyll.

Since V202311, the chlorophyll content of each phytoplankton group is explicitly represented as a model state variable (Geider et al, 1998).

Description of the production chain

The current BS-Biogeochemistry model is online-coupled with the NEMO v4.2 ocean model and is run with a horizontal resolution of approximately 2.5 km and 59 unevenly spaced vertical z-levels. This NEMO configuration includes fine-tuned physical processes critical for simulating Black Sea biogeochemistry, such as the open boundary condition at the Bosporus (based on Stanev et al., 1997; Stanev and Beckers, 1999), strict conservation of the biogeochemical tracer budgets, and the Generic Length-Scale (GLS) turbulence closure scheme. Since V202411, there is feedback from the biology to the spectral light model, which in turn influences water column heating computed in the physical model. BS-BIO NRT system runs daily 1-day analyses and 10-day forecasts.

Once per week (on Tuesdays), a 10-day analysis is performed during which the model assimilates daily L3 satellite chlorophyll observations. The chlorophyll content of the three phytoplankton groups is updated, and the assimilation increment is back-propagated to the carbon and nitrogen content of the phytoplankton groups before the model is restarted. The three phytoplankton groups are modified together in such a way that the ratios between them remain unchanged by the assimilation procedure. Similarly, the ratio between chlorophyll, carbon, and nitrogen in each phytoplankton group is preserved.

Together with surface chlorophyll, the model also assimilates in situ oxygen profiles from BGC-ARGO observations. The assimilation is mono-variate, meaning only the oxygen variable is updated in the model.

The assimilation increments are 3D, i.e., the analysis modifies the model variables also below the surface layer but only down to the bottom of the surface mixed layer.

The present version V202511 model was initialized on 1 January 2022 (from the V202311 simulation).

II.3 Upstream data and boundary condition of the NEMO-BAMHBI model

The CMEMS–BLK- BGC analysis and forecast system uses the following upstream data:

1. **Atmospheric files**: air temperature, air dew point temperature, total precipitation, cloud cover, wind velocity and mean sea level pressure, produced by ECMWF analysis and forecast fields (at 1/8° spatial resolution with 3hr and 6hr temporal resolution). These fields are used to force the NEMO physics and, in the case of wind, to compute the air-sea exchanges of O_2 and CO_2.

2. **Atmospheric deposition of nutrients:** inorganic nitrogen delivered by the CAMS models run at ECMWF. Phosphate deposition is not available from CAMS and is provided by Kanakidou et al. (2012). These atmospheric depositions are of a similar order of magnitude as river inputs and are necessary to sustain primary production in the deep basin.

3. **Satellite data assimilation**: daily L3 Copernicus Marine Environment Monitoring Service (CMEMS) Ocean-Colour satellite observations (product: OCEANCOLOUR_BLK_BGC_L3_NRT_009_151).

4. **In situ data assimilation**: oxygen profiles from BGC-ARGO floats obtained from CMEMS product INSITU_BLK_PHYBGCWAV_DISCRETE_MYNRT_013_034.

5. **Riverine freshwater and nutrient inputs:** The BS-BIO NRT system uses near real-time (NRT) freshwater discharge data from the Romanian Hydrological Institute for the Danube, by far the most important river in the Black Sea. The data is provided for the three main arms of the river. However, the nutrient concentration in river water is still obtained from climatology. For the other rivers (Dniepr, Dniestr, Bug, Rioni, Sakarya, Filyos, Yesil, Inguri, Coroch, Kodori, Bzyband Kizilirmak), freshwater fluxes are obtained from the EFAS platform[5]. Due to the absence of NRT nutrients data, the BS-BIO NRT system uses climatological values for inorganic nutrients and organic matter, computed from long-term time series provided by Ludwig et al. (2009) as part of the EU SESAME[6] and PERSEUS[7] projects.

6. **Bosporus Strait boundary conditions:** The Bosporus is considered as an open boundary (Stanev et al, 1997; Stanev and Beckers, 1999). Current velocity and salinity are set to conserve total seawater and salt in the Black Sea domain. Temperature uses a zero-gradient boundary condition. Biogeochemical variables use zero-gradient ("Neumann") or fixed value ("specified") boundary conditions[8].

7. **Incoming spectral radiation**: Provided by the MAR regional atmospheric model coupled with the ecRAD radiative transfer module (i.e., the same as used in recent ECMWF models; see Hogan, R. J. and Bozzo, A., 2018.

[5] https://european-flood.emergency.copernicus.eu/

[6] https://cordis.europa.eu/project/id/36949/reporting

[7] http://www.perseus-net.eu/

[8] https://zenodo.org/records/6334656

III VALIDATION FRAMEWORK

The quality assessment of the present version V202511 analysis-forecast BS-MFC-Biogeochemistry system (described in the section II.2) was carried out by comparing a simulation with available NRT observations for the period 2022-2023 (i.e., the same validation period used for V202411), allowing comparison of both models' validation.

The product quality assessment combined quantitative metrics and qualitative evaluations. Quantitative validation metrics are reported in **Table 2** and are computed for oxygen and chlorophyll, for which NRT observations are available from satellites and BGC-ARGO. Some variables could not be validated in NRT due to a lack of observations, including nutrients (nitrate and phosphate), phytoplankton biomass, primary production, zooplankton, carbonate system variables and oxygen-demand-units. Some of these variables (nutrients, dissolved inorganic carbon, pH, total alkalinity, pCO2) are validated in the corresponding multiyear product (see the QUID for BLK_MULTIYEAR_BGC_007_005) or, as for carbonate system variables, using historical data of sufficient quality (e.g., Sesame, KNORR). Additionally, density profiles of inorganic variables (i.e., NO_3, O_2, DIC, alkalinity, pH, oxygen demand units (ODU), NH_4, PO_4, SiO_2) are compared with aggregated profiles from the literature.

NRT Datasets

The following datasets were used to validate the product:

1) In situ ARGO observations (from the CMEMS in-situ TAC product: INSITU_BLK_PHYBGCWAV_DISCRETE_MYNRT_013_034[9]; also obtainable from Coriolis[10]:). They are used for the validation of physical variables such as light, temperature, salinity, and derived quantities including MLD, cold intermediate layer (CIL), cold content (CCC)) as well as biogeochemical variables (oxygen, chlorophyll).
   Chlorophyll estimates from BGC Argo fluorescence measurements are corrected to account for the Black Sea specific conditions (higher content of CDOM, anoxia) as described in Ricour et al. (2021).
2) BLKSEA_ANALYSISFORECAST_BIO_007_010 V202211 to V202411 (i.e., the previous versions of the product object of this document)
3) CMEMS OC-TAC satellite level-3 observations for the Black Sea were obtained from CMEMS products[11] OCEANCOLOUR_BLK_BGC_L3_NRT_009_151 and OCEANCOLOUR_BLK_BGC_L3_MY_009_153 (plankton, transparency, reflectance).

Metrics

The quality assessment uses statistics that quantify different aspects of model-data mismatches (**Table 2**, further details in Allen et al., 2007): bias and percentage bias, standard deviation, Pearson correlation, Nash-Sutcliffe efficiency, reduced chi-square In addition to these global statistics, the model ability to simulate the monthly vertical profiles of oxygen and its maximum value is assessed.

For the validation of the chlorophyll variable, the model – observation comparison has been performed independently in 11 sub-regions of the Black Sea. The subdivision is done on the basis of the bathymetry

[9] https://doi.org/10.48670/moi-00033

[10] http://www.coriolis.eu.org/ (last accessed 06/11/2024)

[11] https://doi.org/10.48670/moi-00301

(<50 m, 50 to 200 m, and > 200 m,see Kopelevich et al., 2013). Furthermore, each sub-region is divided into eastern and western zones. The map of all sub-regions is shown in **Figure 1**.

Table 2: List of metrics used to assess model performance. For the definition of the metrics, "O" represents the observations and "P" is the model output. All metrics listed in the table are dimensionless.

Definition	Meaning	Interpretation
Percentage bias $PB = \frac{P-O}{O} * 100$	Compares the mean values of *O* and *P*, and expresses the average bias as a percentage value of the *O* average	PB > 0: global overestimation PB < 0: global underestimation |PB| < 10%: excellent skill, 10-20: very good, 20-40: good, > 40 poor (Maréchal, 2004)
Standard deviation ratio $r_\sigma = \frac{\sigma_O}{\sigma_P}$	Compares the distribution of *O* and *P* around their respective averages. s represents the standard deviation	> 1: higher variability in observations < 1: higher variability in the model
Nash-Sutcliffe efficiency $N - S = 1 - \frac{\sum(O-P)^2}{\sum(O-\overline{O})^2}$	Relates the variance of the difference between model predictions and observations to the variance in the observations $\sum$ represents the sum operator over the respective dataset.	> 0.65: excellent > 0.5: very good > 0.2: good < 0.2: poor (Maréchal, 2004)
Pearson correlation coefficient ρ	Indicates how variations of *O* and *P* around their respective means are related	
Reduced chi-squared statistic $\chi^2 = \frac{1}{n\,\sigma_O^2} \sum (P-O)^2$	Ratio of squared model-observation differences, and observation variance	We use this statistic for comparing different versions of the model (model evolution).

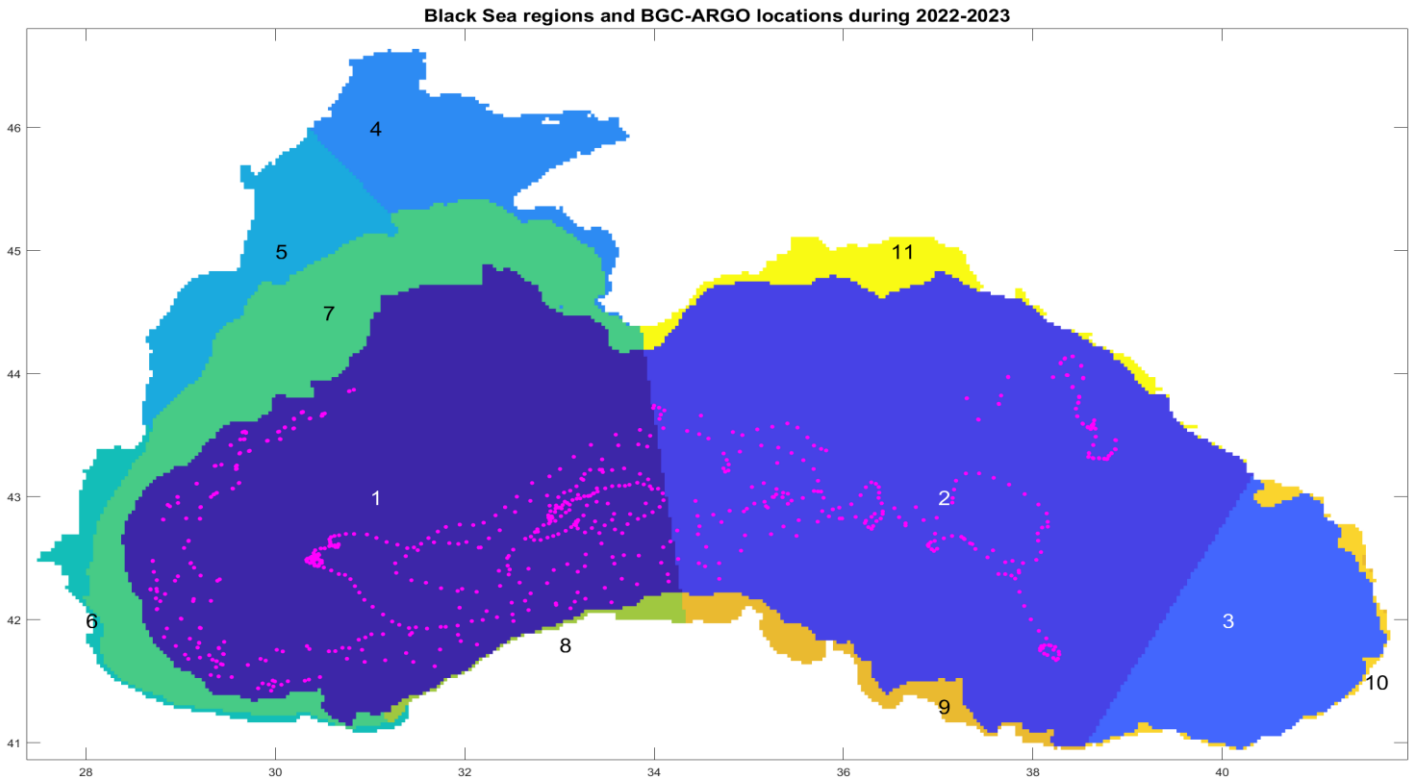

***Figure 1**: Black Sea sub-regions (based on Kopelevich et al., 2013) used for computing the chlorophyll quality metrics. Numbers indicate the regions of the Black Sea. Regions 1-3 are "open sea", regions 4-5 represent the North-Western Shelf, and region 7 is a transition region. Regions 6, 8-11 represent narrow coastal regions. Pink dots represent the locations of collected Argo profiles.*

IV VALIDATION RESULTS

In the following subsections, we first analyse the performance of the BLKSEA_ANALYSIS_FORECAST_BIO_007_010 product at the V202511 release, focusing on:

- **Physical variables** that are known to act as primary control in biogeochemistry: mixed layer depth, "Cold Intermediate Layer" cold content.
- **Biogeochemical variables**, focusing on the validation of oxygen, chlorophyll, PAR and Kd, because these are the only variables available in NRT.
  - For oxygen, we assess global statistics of the model (computed over the whole model domain and simulation period), its ability to reproduce monthly oxygen vertical profiles, and value of the maximum concentration.
  - For chlorophyll, we compare model results with satellite products for the different regions identified in **Figure 1** and with the chlorophyll retrieved from fluorescence data provided by BGC-ARGO.
  - Finally, the carbonate system was qualitatively compared to climatology.

IV.1 Physical variables

The Black Sea presents an oxygen-rich surface layer overlying an anoxic water mass. Basin ventilation is mainly governed by the annual formation of cold oxygen-rich waters that accumulate at depth to form the Black Sea Cold Intermediate Layer (CIL). This ventilation mechanism is limited by the presence of a permanent halocline. In this section, we assess the capacity of the BS-BIO physical model (NEMO 4.2) to simulate the formation of the CIL and the associated mixing processes by comparison with observations.

IV.1.1 Mixed layer depth (MLD)

In the Black Sea, the typical criterion to define MLD is usually the depth at which density is 0.125 $kg.m^{-3}$ llarger than the reference density at 3 m depth (Kara et al., 2009, and references herein). However, this definition is limited by the lack of high-quality data close to the surface (< 10 m depth). Hence, the model MLD is computed at each point as the depth where the density exceeds the density at 10 m by 0.125 kg m^{-3}, and then spatially averaged over the basin. The basin averaged model MLD (for points where the depth is at least 50 m) is compared with MLD derived from individual Argo profiles.

The model is able to reproduce the typical seasonal MLD cycle (**Figure 2**) and, specifically, in the present version V202511, the model simulates realistic MLD for all seasons.

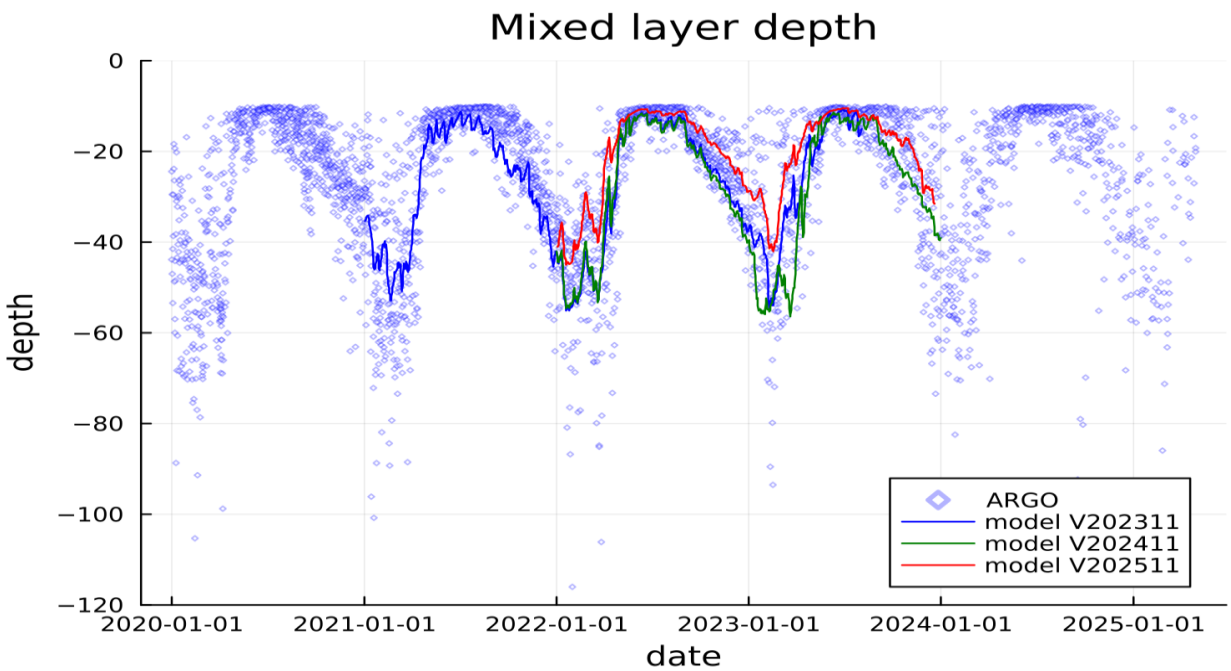

Figure 2. *Modelled mixed layer depth (MLD) spatially averaged over model estimates where the bathymetry is deeper than 50 m (red line) compared to MLD calculated from individual ARGO T,S profiles (blue symbols).*

IV.1.2 Cold intermediate layer

The Black Sea Cold Intermediate Layer (CIL) is characterized by temperatures below 8.35°C and densities greater than 1014 kg m^{-3}. This water mass has been extensively described in the literature (e.g., Capet et al., 2015, Miladinova et al., 2018). The amount of CIL, referred to as "CIL cold content" (CCC) is estimated from temperature and density data as the heat deficit of the CIL relative to a 8.35°C reference layer. **Figure 3** shows the CCC estimated by NEMO model versions V202105 to V202511.

For previous versions, the agreement between model results and observations has been demonstrated in previous versions of this QUID. With the upgrade to V202511, the CCC formed during the winters is smaller than in previous model versions. All NEMO versions show that the CCC is smaller during the present decade compared to previous decades. Furthermore, during the present decade, the CIL has been destroyed more rapidly. In particular, since around 2020, the CIL formed during winter disappears completely in the following spring. Although during winter 2022 slightly more CIL was produced than in the previous years, the mild winter in 2023 resulted in no CIL creation at all.

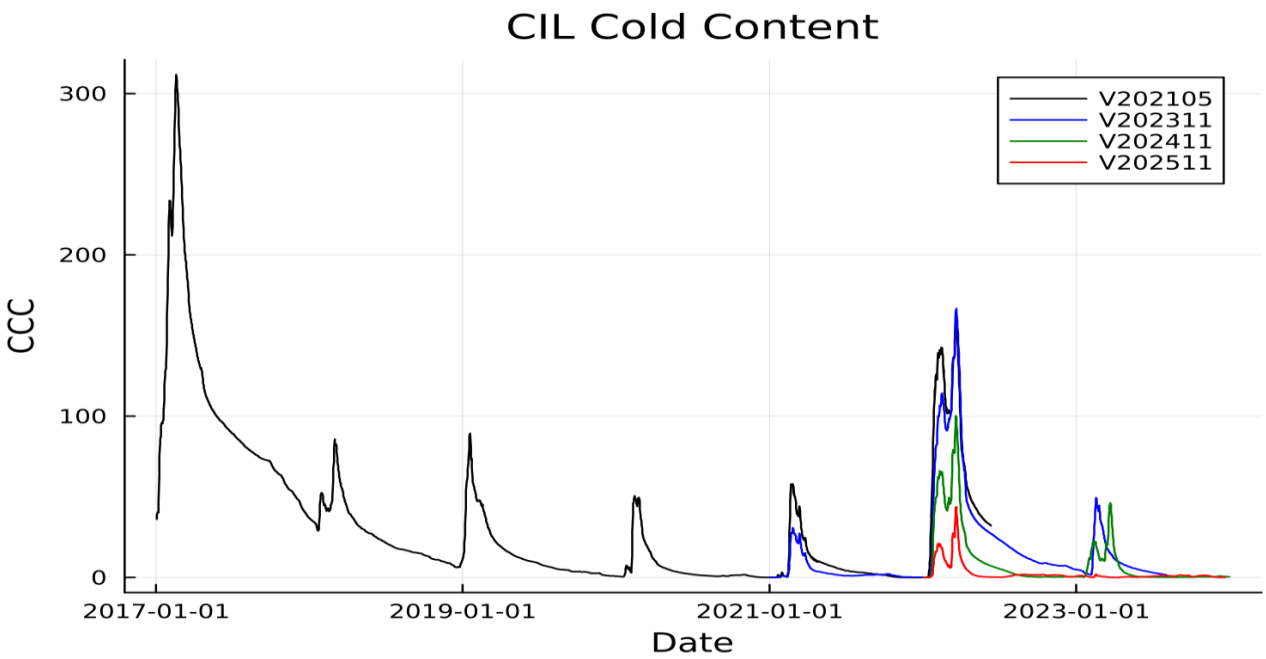

Figure 3. *Cold Intermediate Layer (CIL) Cold Content in the Black Sea from the older (V202105) up to present (V202511) version.*

IV.2 Oxygen

The Black Sea's oxycline had shoaled in the last decades (Capet et al., 2016a). The free model successfully reproduces this shoaling, as demonstrated in the previous versions of this product and in the MY product BLKSEA_MULTIYEAR_BGC_007_005. Since version V202411 of the model, the BGC-ARGO oxygen observations are assimilated. BGC-ARGO cycles last five days in the Black Sea. The validation metrics obtained below are computed from model forecasts (*prior to* assimilation of BGC-ARGO oxygen observations), but even so, the oxygen data are not fully independent, as they are obtained from the same buoys that provided the observations for the validation, as they were assimilated during the previous cycle (i.e., 5 days earlier).

The error metrics are improved compared to previous model versions. Table 3 presents model-ARGO comparison with a bias of only2.3 mmol.m^{-3}.

Table 3: *Model validation metrics for oxygen based on all model prediction-observation pairs with respect to BGC-ARGO observations (for locations see* ***Figure 1****). Validation metrics are defined in* ***Table 2****.*

Oxygen	Unit	Value at V202311	Value at V202411	Value at V202511
Bias	mmol m^{-3}	9.28	2.3	**1.62**
RMSD	mmol m^{-3}	31.75	17.4	**15.9**
Standard deviation ratio	dimensionless	0.82	0.92	**0.93**
Nash-Sutcliffe	dimensionless	0.85	0.93	**0.94**
Pearson coefficient	dimensionless	0.96	0.97	**0.97**

During January 2022 – December 2023, a total of 720 BG-ARGO profiles with oxygen data were collected, all from the deep-sea. Some profiles contained suspicious data (negative oxygen concentrations) and were rejected. Figure 4 compares the remaining BGC-ARGO oxygen profiles with the model, arranged by month (approximately 50 profiles per month).

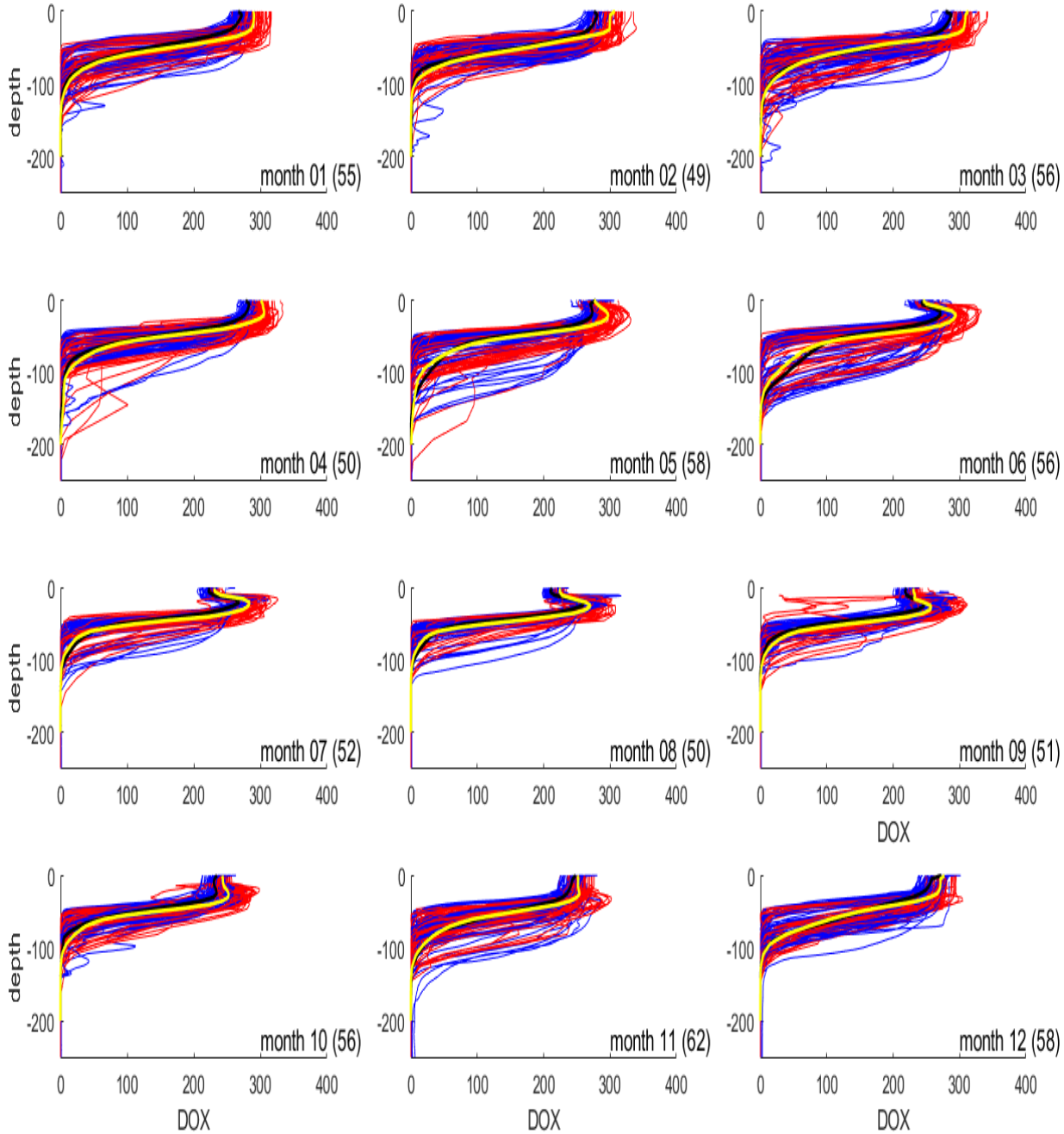

***Figure 4**: Oxygen profiles for each month from January 2022 to December 2023 (model in red, in situ observations in blue; the model and observed monthly mean profiles are respectively in yellow and black). For each month, the number of profiles is specified in parentheses.*

The depth of the oxygenated layer is well simulated by the model. The bias is 8.03 m (unchanged from V202411), and the Pearson correlation coefficient is 0.81. Figure 5 shows the depth at which the oxygen concentration falls below 20 mmol.m^{-3} for both the model and the observations.

Similar to previous versions, in the present version of the product the oxygen maximum is accurately simulated by the model, with a mean bias of 16.3 mmol m^{-3}, corresponding to a percentage bias of 5.9 % (a very small degradation compared to V202411). As an example, the complete timeseries of oxygen measurements from BGC-Argo buoy 3902005 during 2022-2023, along with the corresponding model values are shown in Figure 6.

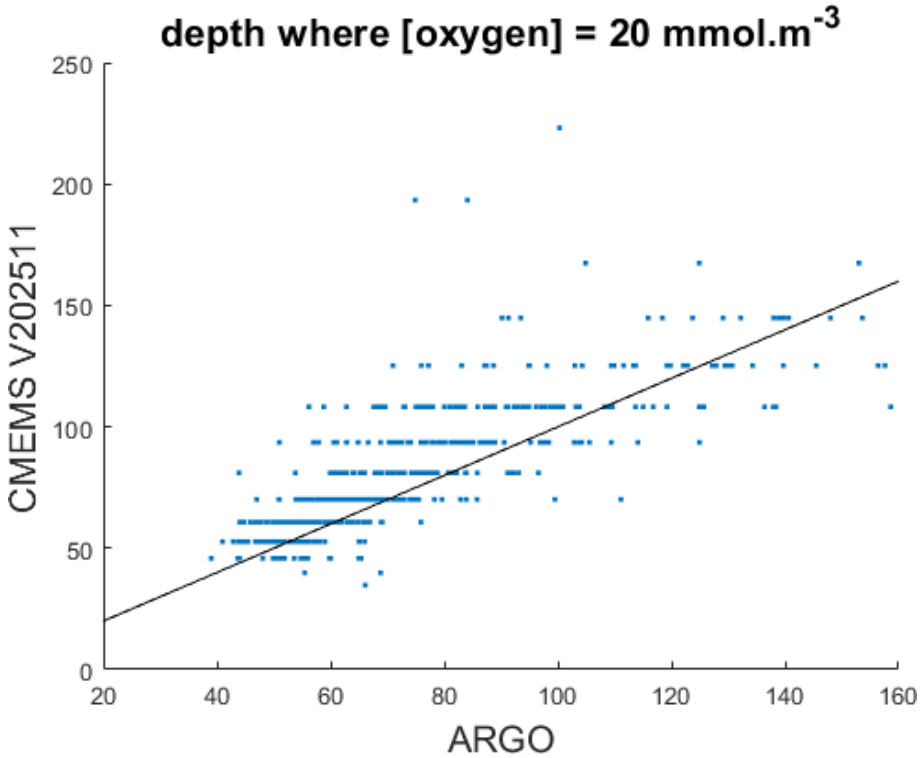

***Figure 5**. Depth at which oxygen falls below 20 mmol.m-3 for the BGC-ARGO (x-axis) and the model (y-axis). The identity line is plotted in black for reference.*

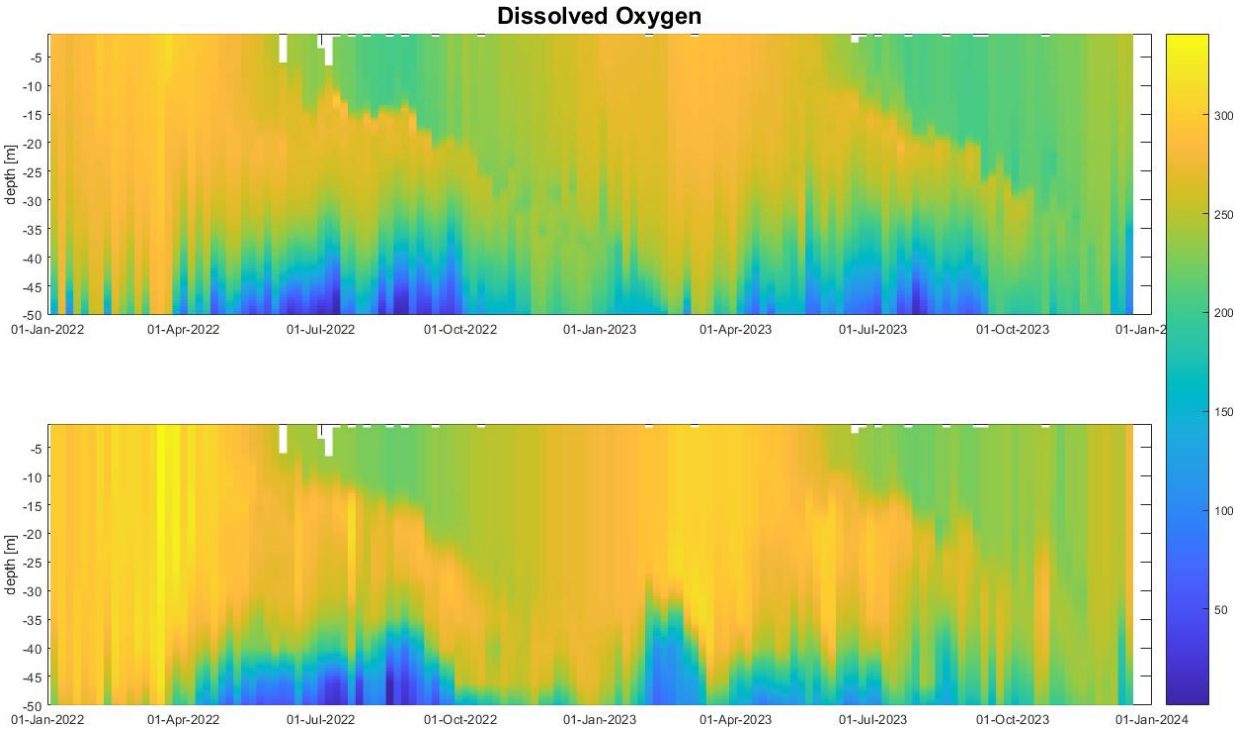

Figure 6. *Model-observation comparison for oxygen: Top: Hovmöller diagram from BGC-ARGO buoy 3902005 during 2022-2023. Bottom: model equivalent. Oxygen units are mmol.m-3.*

IV.3 Chlorophyll

In this section, modelled chlorophyll concentrations are compared to satellite L3 chlorophyll concentrations and BGC-ARGO float data for the period 2022-2023.

The switch to V202211 introduced a higher vertical resolution, allowing the representation of the shallow summer thermocline, which sustains a second deep chlorophyll maximum. Furthermore, the use of near real-time (NRT) freshwater discharge data for the Danube improved the representation of chlorophyll concentration (e.g., enabling the identification of a late-summer/autumn bloom that was absent in previous model versions). These modifications are still present in V202511. Furthermore, NRT freshwater discharge data is now also used for the other rivers (see Section III Validation framework). In V202511, atmospheric nitrogen deposition is also more realistic, as it is provided from a real-time dataset (CAMS) rather than climatology.

IV.3.1 Satellite data

The regional time-series consist of daily averages of all available data in each sub-region (**Figure 1**). For satellite products, if the level-3 (L3) image of a particular region contains too many clouds and has fewer than 100 valid pixels, then the image for that day and region is omitted from the time series (leaving a gap).

The satellite observations used in the comparison are taken from the CMEMS MY reprocessed L3 images (CMEMS product 009_153[12], reprocessed product based on multi-satellite composite images). These images are generated using an algorithm that combines results from a neural network for coastal areas with a classical ratio-based method for the open sea.

Timeseries of model and observations are shown in Fig. 7. In the open sea, the bias of the model for chlorophyll values with respect to satellite estimates is -0.0089, -0.0186, and 0.0188 $mg.m^{-3}$ in the western, central, and eastern basins, respectively. The RMSD is 0.07 $mg.m^{-3}$, and the standard deviation ratio is between 1.04 and 1.1, indicating that the model has a slightly larger standard deviation than the satellite estimates. The Pearson correlation coefficient is 0.95 in all three regions, and the Nash-Sutcliffe coefficient is “excellent” (> 0.75).

In region 4 (influenced by the Dnestr and Dnepr rivers, see Fig. 1) and 5 (directly in front of the Danube), nutrient discharge by rivers is critical for chlorophyll predictions. Although freshwater discharge is obtained in near-real time, the nutrient concentrations in this freshwater are based on a monthly climatology. Hence, in these regions (4 and 5), some differences between model and observations persist even after the assimilation of the satellite chlorophyll. The resulting bias is positive: 0.44 and 1.10 $mg.m^{-3}$ in regions 4 and 5, respectively.

In region 6, which is also a shelf region located South of the NWS but less influenced by rivers, the bias is negative and small: -0.07 mg m^{-3}. Region 7 is the transition region between the NWS and the open sea, and here also, the model and observations are very close (bias = 0.018 mg m^{-3}).

Table 4 shows the bias between model and observations, expressed in $mg*m^{-3}$. The error statistics indicate a similar very low bias, which is a consequence of the improved chlorophyll representation in the model, and the use of data assimilation.

[12] https://doi.org/10.48670/moi-00303

Table 4: *Bias for chlorophyll obtained for the previous version (V202411) and the current version (V202511), computed from the average of the observation-model prediction pairs for the different regions (1 to 11). The bias accuracy metric has been defined in Table 2. Observations are from the near-real time CMEMS satellite products.*

EAN \ Region	1	2	3	4	5	6	7	8	9	10	11
Bias V202411	0.0013	-0.018	0.011	0.65	0.25	-0.006	-0.001	0.101	0.062	0.219	-0.053
Bias V202511	**-0.0089**	**-0.019**	**0.019**	**0.44**	**1.10**	**-0.070**	**0.018**	**0.048**	**0.054**	**0.359**	**-0.060**

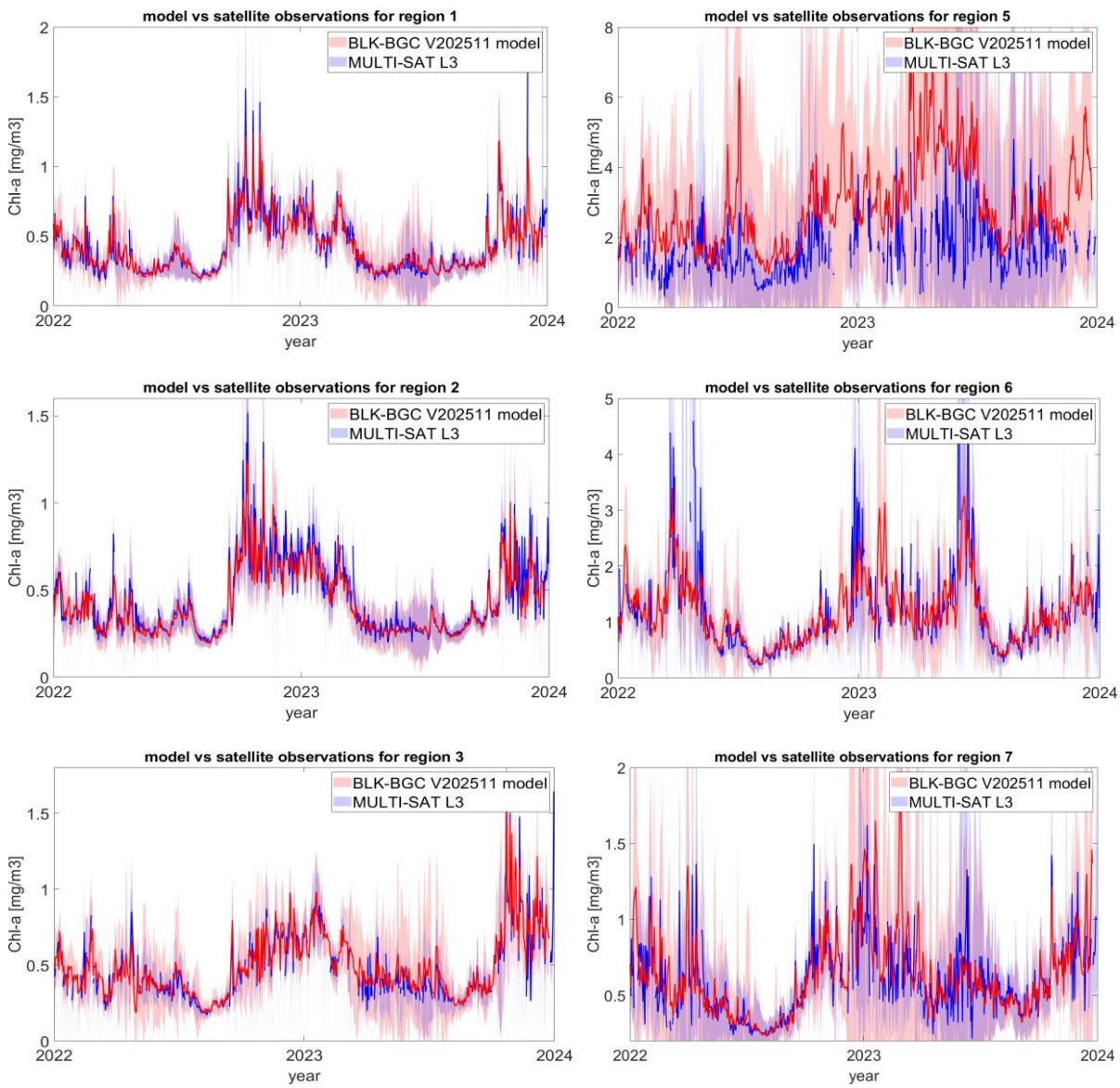

Figure 7: *Time series of satellite (in blue) and model (in red) chlorophyll spatial averages computed for (left column) open sea sub-regions 1, 2 and 3, and (right column) NWS regions 5, 6 and 7. The regions are shown in Fig 1. The shaded area represents ± 1 standard deviation around the regional mean.*

IV.3.2 In situ data

During the validation period, one or two BGC-ARGO floats were measuring fluorescence. However, in the deep and anoxic part of the Black Sea, chlorophyll concentration cannot be directly estimated from BG-ARGO fluorescence data, as anoxic conditions prevent efficient chlorophyll degradation, causing fluorescence to increase with depth. With generic algorithms, this would lead to erroneous results (e.g., chlorophyll concentrations of 0.3 mg/m^{-3} at 1000 m depth). Therefore, a specific correction (based on the model of Xing et al., 2017) was applied to essentially set deep chlorophyll to zero (Ricour et al., 2021).

Regarding the DCM, the model shows higher strength than observed (bias = 0.43 mg m^{-3}) and somewhat lower variability (standard deviation ratio = 0.77). Its depth is overestimated, with a bias of 23.89 m (Table 5). Some error statistics are slightly improved compared to V202411, while others decrease marginally.

Table 5: *Validation metrics (see Table 2) for chlorophyll estimates based on pairwise comparisons with in situ ARGO observations (see **Fig. 1**).*

Quantity	**Intensity of chlorophyll maximum**
Bias (mg.m^{-3})	**0.43**
RMS (mg.m^{-3})	**1.31**
Standard deviation ratio r_σ (/)	**0.77**
	Depth of chlorophyll maximum
Bias (m)	**23.89**
RMS (m)	**27.29**
Standard deviation ratio r_σ (/)	**1.12**

IV.4 Carbonate sub-system

The validation of the carbonate system is not possible in NRT. Besides, even for the past, the quality of carbonate data (pH, alkalinity) is sometimes not optimal. Also, for assessing the consistency of the simulated carbonate components, we used KNORR 88 observations described in Goyet et al. (1991), as well as Sesame 2008 data (Borges, personal communication). In addition, we conducted comparisons with typical profiles collected on a density scale and presented in Moiseenko et al. (2011). When comparing model values with climatologic measurements, it is preferable to use the density anomaly, σ_t [kg m^{-3}] instead of depth, because in the Black Sea most biogeochemical variables exhibit distinct characteristics at specific density levels. At the surface, σ_t is around 11, increasing to 14-16 in the oxycline zone, before reaching 16.2 at the top of the anoxic layer and then 17-17.5 at the bottom.

IV.4.1 Oxygen and Nitrate

The simulated oxygen on a density scale is shown in **Figure 8**. Oxygen in the deep part of the Black Sea was already compared to observations in Section IV.2. The model successfully reproduces the disappearance of oxygen at densities of 15.8-16 kg.m^{-3}, in agreement with observations. In **Figure 9**, the nitracline is clearly visible around σ_t=15.2 kg m^{-3} with a maximum nitrate concentration of about 5 to 6 mmol m^{-3}. As in V202411, the nitracline vanishes at σ_t=16.0-16.5 kg m^{-3}, whereas in V202211 it extended deeper (up to σ_t=17 kg m^{-3}). Surface values are highly variable in time and space. On the NWS, very high values can be found close to river mouths (blue points in Figure 9). These results are largely consistent with V202411.

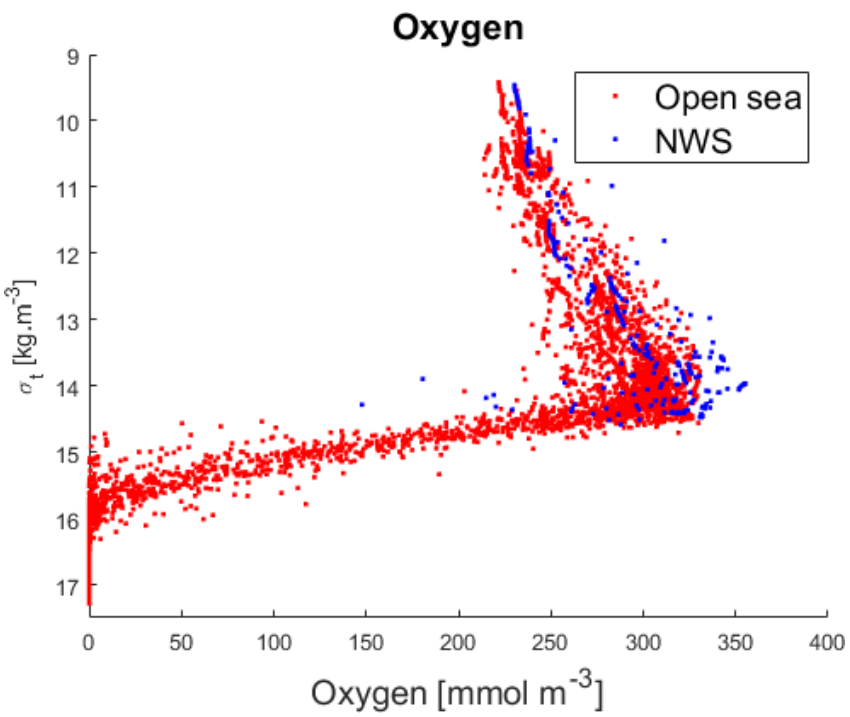

Figure 8: *Oxygen as a function of the density anomaly. Vertical profiles are selected randomly in space, and throughout the simulation period (2022-2023) from V202511 model results.*

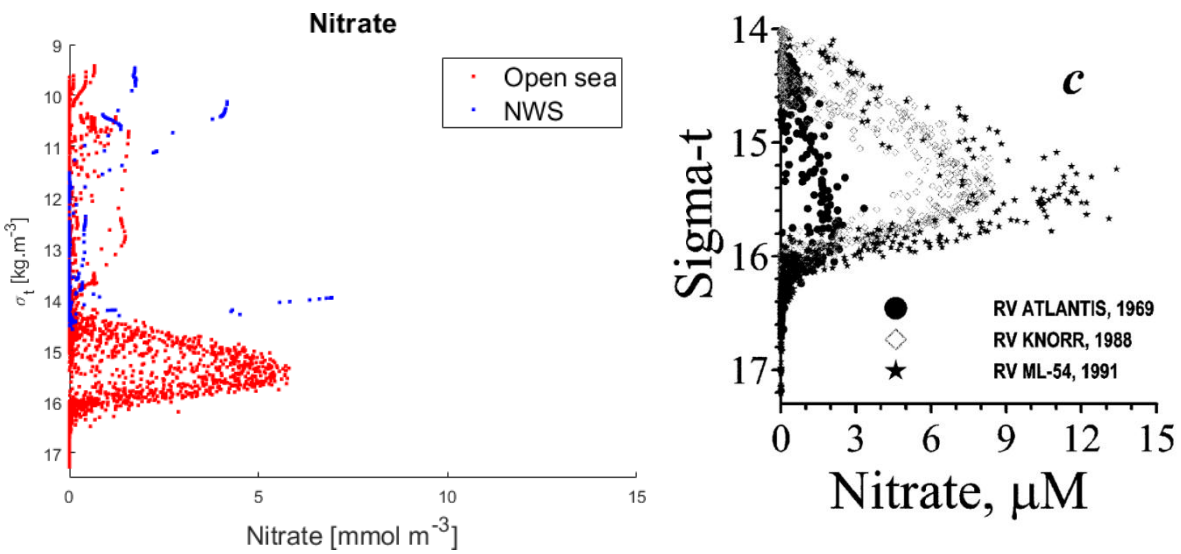

Figure 9: *Nitrate vs density anomaly. Left panel: CMEMS BS-MFC V202511 model (2022-2023). Right panel: reprinted from Konovalov and Murray, 2001, with kind permission from the authors. Compared to the eutrophication period 1988-1992, the nitrate peak has decreased to ~4-6 µM (Borges, 2008, unpublished Sesame data in the deep basin) which is of the same order of magnitude as model results.*

IV.4.2 Dissolved Inorganic Carbon

For the considered period (2022-2023), there are no available measurements of DIC, pH, total alkalinity, $spCO_2$, and air-sea CO_2 flux. We therefore compared the model results to values from the literature (**Figure 10**, **Figure 11**, **Figure 12**, **Figure 13**). The modelled DIC is in good agreement with literature values (**Figure 10**).

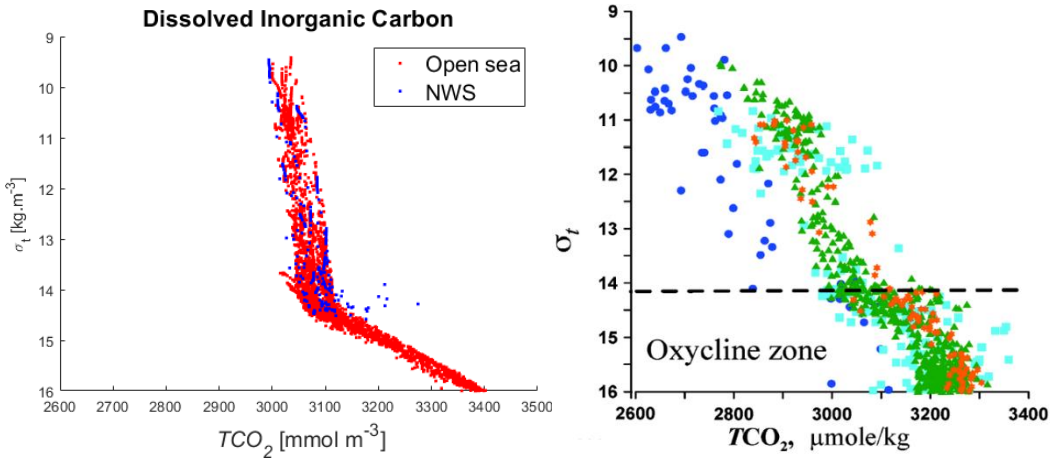

Figure 10: *Dissolved Inorganic Carbon as a function of the density anomaly. Left panel: CMEMS BS-MFC V202511 model (2022-2023). Right panel: reprinted from Moiseenko et al., 2011.*

Also modelled pH and alkalinity are in good qualitative agreement with literature values (**Figure 11**).

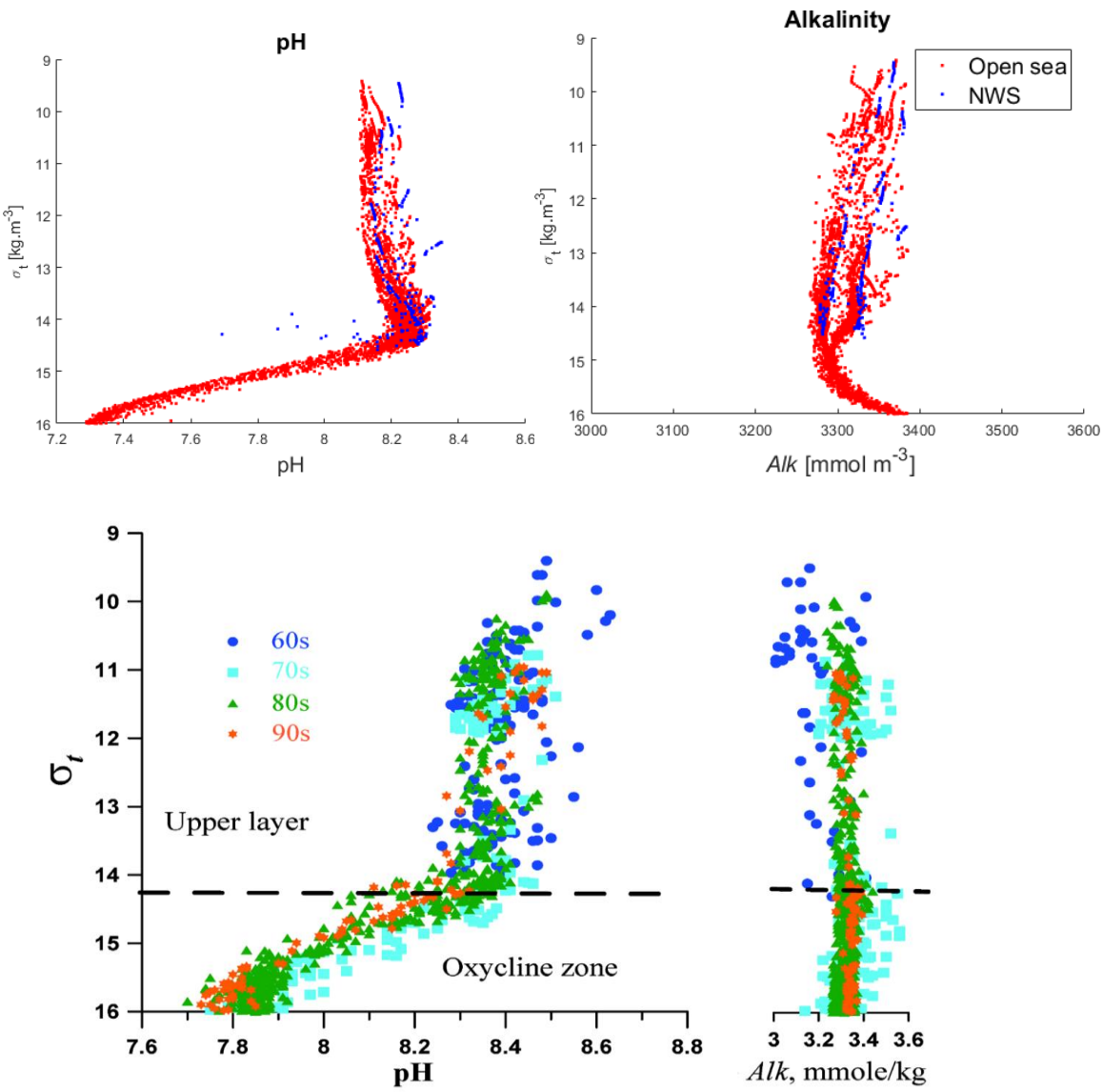

Figure 11: *Comparing model results and observations of pH (left) and Alkalinity (right) as a function of density anomaly (2022-2023). Top row: model predictions. Bottom row: reprinted from Moiseenko et al. (2011), colours indicate the decades (1960s, 1970s, 1980s, 1990s).*

IV.4.3 pC02

For pCO_2, the model shows good qualitative agreement with literature values (**Figure 12**). Clearly, pCO_2 has increased since the 1960s (right panel) and model estimates (left panel) match well the higher values observed in more recent decades.

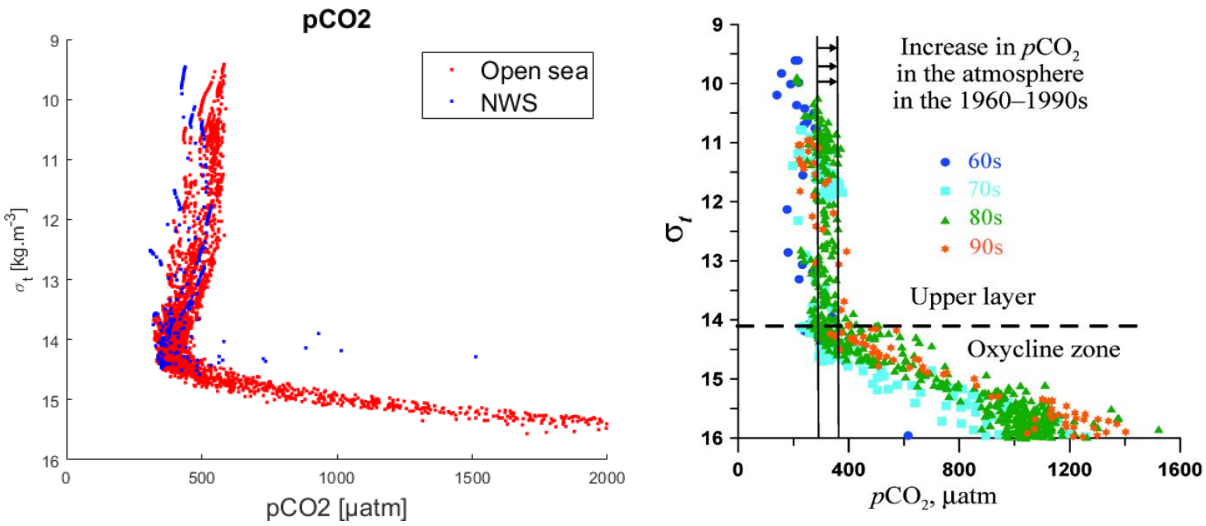

Figure 12: *Comparing model results and observations of pCO_2 as a function of density anomaly. Left panel: CMEMS BS-MFC V202411 model (2022-2023). Right panel: observations, reprinted from Moiseenko et al. (2011), colours represent decades.*

IV.4.4 Air-Sea CO2 Flux

Time-series of spatially integrated air-sea CO_2 fluxes are shown in Fig 13. Due to the lack of observational data, model predictions in region 4 are compared to literature values for the period 2011-2014 (Khoruzhiy, 2018). The modelled values fall within the range report by Khoruzhiy (2018), who observed average fluxes between 6 mmol and -19 mmol $CO_2.m^{-1}.atm^{-1}.day^{-1}$. The seasonal variability is of comparable magnitude in both model and observations.

Since V202311, the model uses time-varying atmospheric pCO_2 with seasonal and interannual variations (e.g., ~433-444 ppm in 2025).

In V202511, the annual mean CO_2 flux in the open sea is upward (i.e., the deep sea is a source of atmospheric CO_2). On the NWS and all the coastal regions, the annual mean CO_2 flux is downward, with the highest values occurring in region 5, which is influenced by the Danube river.

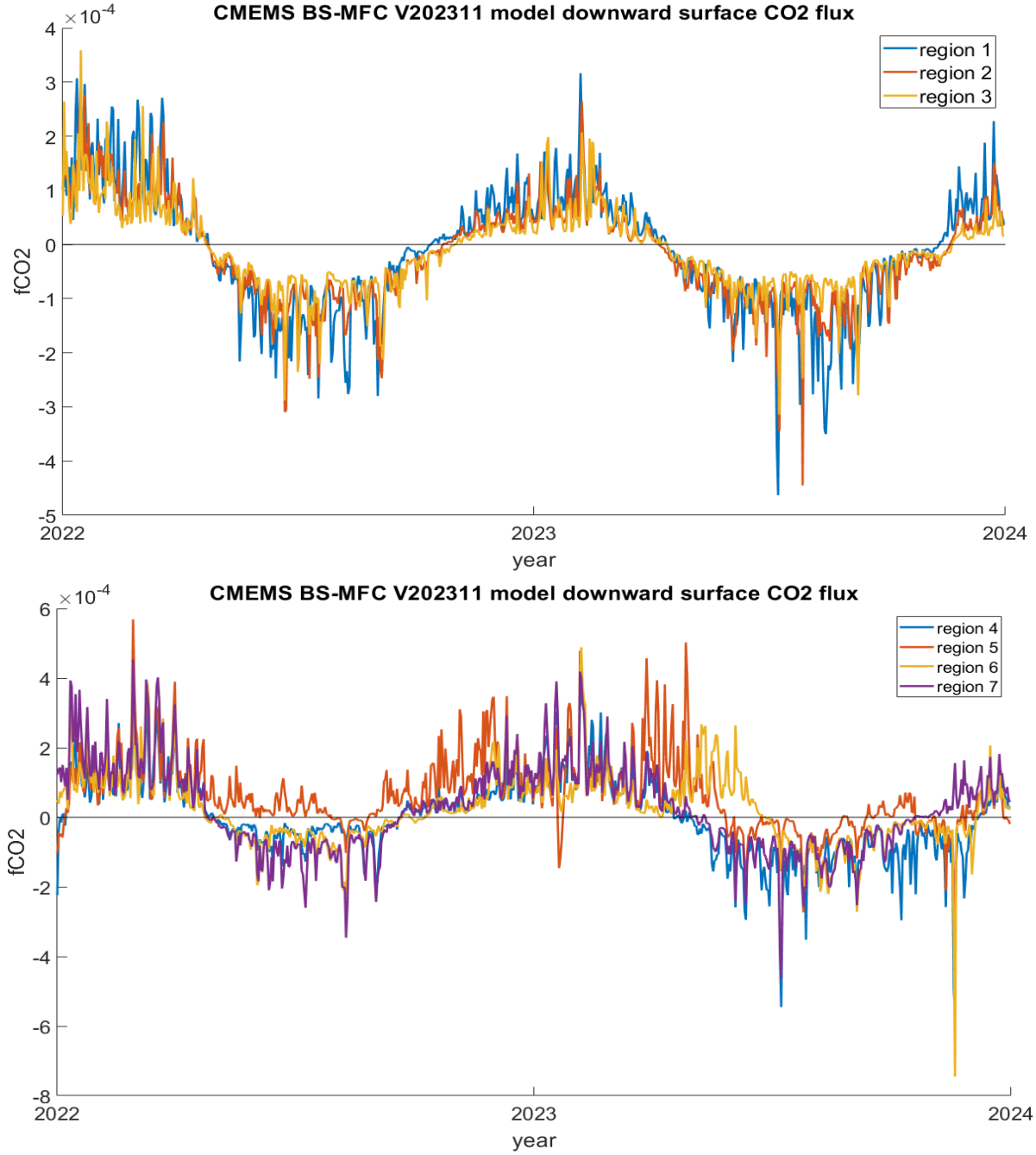

Figure 13: *Time-series of modelled air-sea CO_2 flux over the open sea regions (upper panel) and NWS regions (lower panel). The fluxes are positive downward and are given in mmolC m^{-2} s^{-1}.*

IV.5 Optics

NEMO-BAMHBI-RADTRANS receives incoming spectral radiation at the sea surface from the Regional Atmospheric Model (MAR), which is forced by ECMWF fields at the lateral boundaries. The MAR model is coupled to the ecRAD atmospheric spectral model (the same model used also at ECMWF, see Hogan, R. J. and Bozzo, A., 2018.

Light in each spectral band reaching the sea surface is then propagated to depth. For this aim, the absorption, scattering and back-scattering (both by seawater and by optically-active biogeochemical components) are taken into account.

The validation of the optical product is performed using (1) instantaneous measurements of PAR and irradiation (at a wavelength of 490nm) from BGC-Argo floats, and (2) remote-sensed surface reflectance, and satellite estimates of optical attenuation coefficients (Kd_{490}).

IV.5.1 Photosynthetically Active Radiation (PAR)

The model delivers daily-averaged profiles of PAR while BGC-ARGO delivers instantaneous values. Therefore, PAR profiles are normalized (using the surface value) for the comparison. Surface PAR is estimated by fitting an exponential curve to each observed profile. Observations collected with over 95% of total cloud cover were rejected, as low-light conditions may result in low-quality observations. Cloud-cover was estimated from the corresponding variable in the ECMWF IFS product used as atmospheric forcing by the model.

As there is essentially no light penetrating below 75 m, EANs are calculated for both the full water column and the upper 75 m (Table 6). All EANs are slightly improved compared to V202411.

Table 6: *Model validation metrics for normalized PAR based on pairwise comparisons with BG-ARGO profiles (locations are given in Fig 1).*

PAR/PAR_0	**all observations**	**upper 75 m**
Bias [/]	0.005 m^{-1}	-0.021
Percentage bias [%]	5.85 %	-14.03 %
Rms [/]	0.11 m^{-1}	0.12
Nash-Sutcliffe (dimensionless)	0.07	0.72
Pearson coefficient (dimensionless)	0.28	0.85

A summary of all normalized PAR profiles is shown in Fig 14. There is a small negative model bias visible up to 10 meters depth, approximately. All profiles for the drifting buoy 6903240 (which drifts around the whole deep part of the Black Sea, everything except the north-western shelf and the coastal areas) are shown in Fig 15 as example.

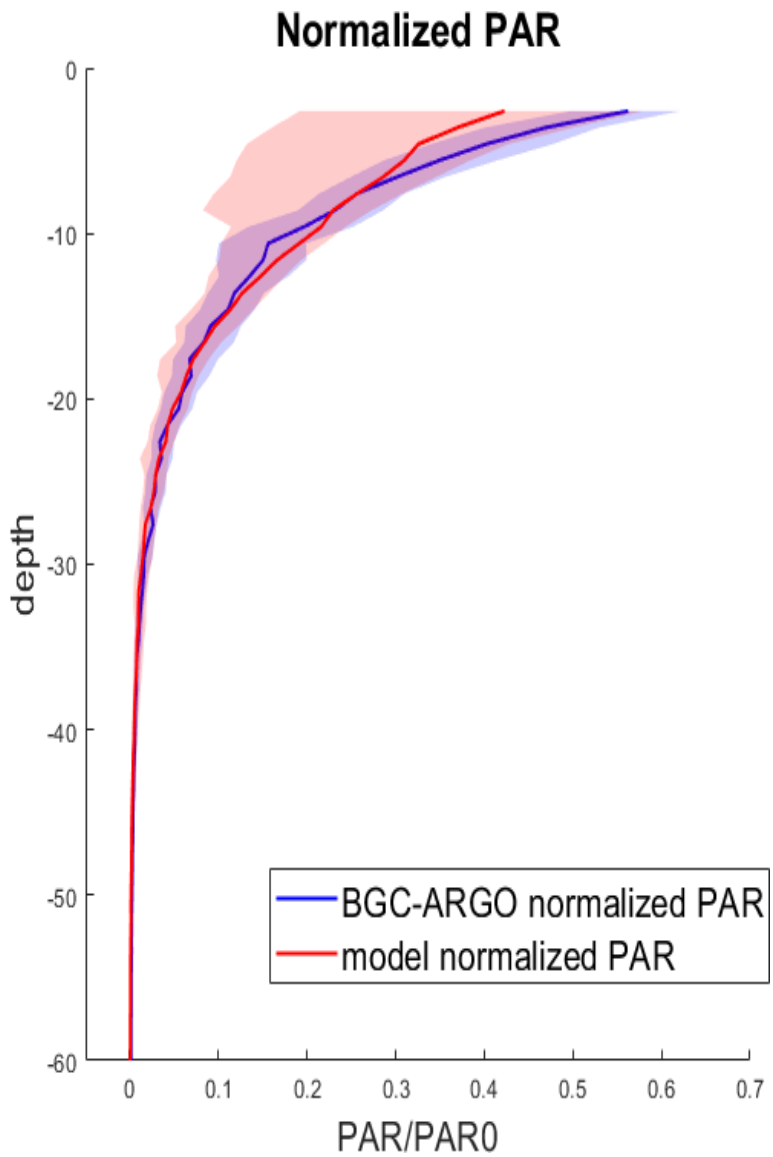

Figure 14: *Normalized PAR (/) as a function of depth (m). Blue: BGC-ARGO observations over 2022-2023. Red: corresponding model results. Solid lines represent the median, while shaded areas cover percentiles 25% to 75%.*

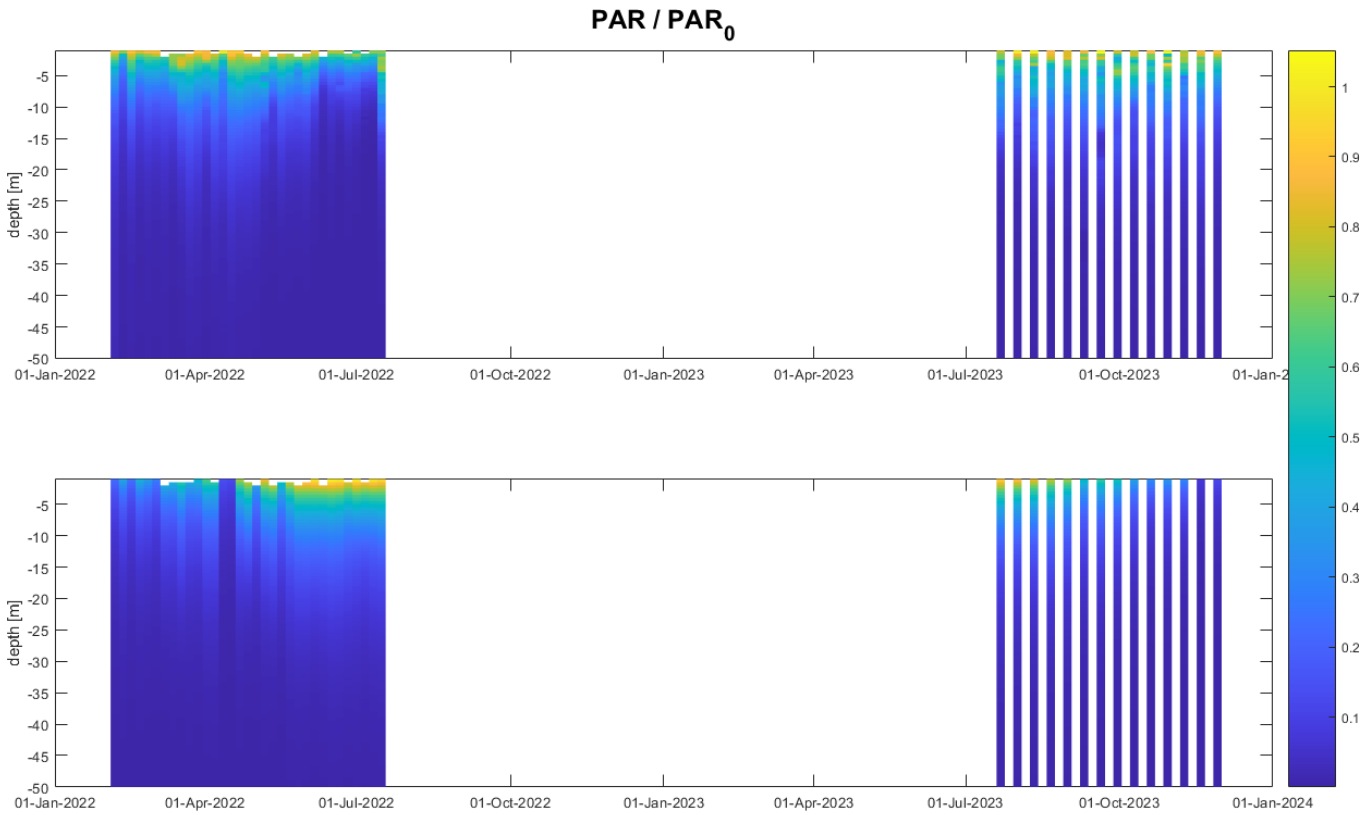

Figure 15: *Hovmöller diagrams of normalized PAR during 2022-2023. Upper panel: BGC-Argo measurements from buoy 6903240 (up to 07/2022) and buoy 7901065 (after 05/2023). Lower panel: model at V202511.*

IV.5.2 Optical attenuation coefficient (Kd)

The model attenuation coefficients across 33 wavebands account for attenuation by both seawater and the various optically active biogeochemical components.

The BGC-Argo attenuation coefficients are derived by fitting exponential curves to downwelling irradiance profiles. These irradiance profiles are measured at different discrete wavelengths. Similarly, surface attenuation coefficients from satellite observations at different wavelengths, are not directly measured but derived at discrete wavelengths from reflectance data.

For satellite measurements, we use Kd(490 nm) delivered by the multi-satellite product OCEANCOLOUR_BLK_BGC_L3_NRT_009_151, which is directly equivalent to the model variable.

Ocean colour satellite products represent conditions in the optical layer, typically defined as the 1/Kd depth. In this report, the model - satellite comparison is performed over the first 5m of the water column.

Fig 16 shows a comparison for different regions (see Fig 1) between satellite-derived Kd(490) and modelled Kd(490), horizontally averaged over the surface layer. Bias is around -0.02 m^{-1} in the three deep ocean regions, (unchanged compared to V202411), and around -0.03 m^{-1} in the other regions, except for regions 5 and 6, where the bias is around -0.06 m^{-1}. As the bias is negative, the model tends to underestimate Kd. As for PAR, the estimation of Kd is less accurate in the NWS than in the open sea, but this underperformance is significantly reduced compared to model versions before V202411.

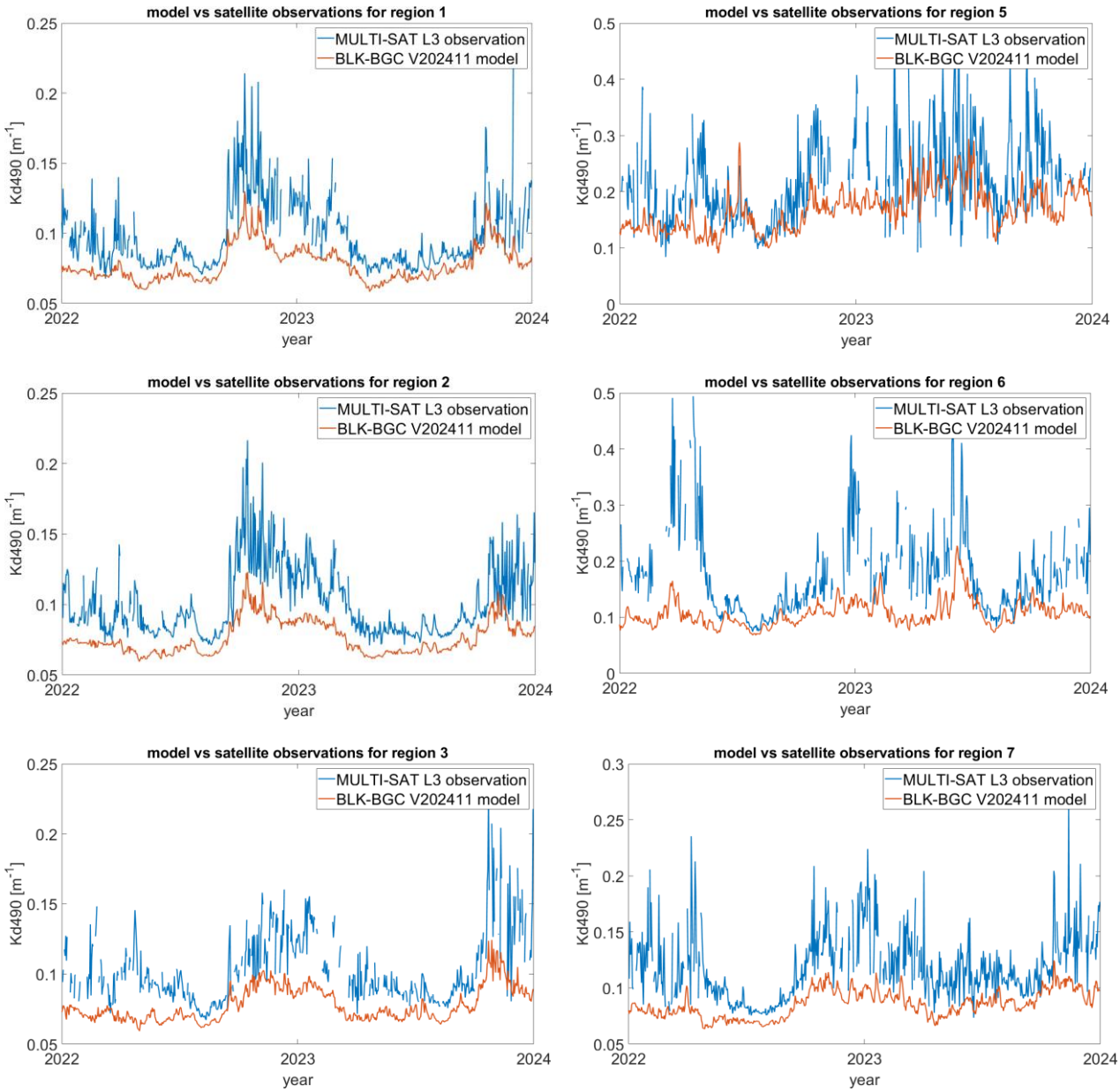

Figure 16: *Time series of satellite (blue) and model (orange) Kd spatial averages computed for (left column) open sea sub-regions 1, 2 and 3, and (right column) NWS regions 5, 6 and 7. The regions are shown in* ***Figure 1****.*

Kd(490) from BGC-Argo is estimated by first smoothing the downwelling irradiance delivered by Argo at 490 nm and then fitting a decreasing exponential function at each observed depth level. Without smoothing, the obtained attenuation profile would be extremely noisy, with fluctuations exceeding 100% between adjacent depths.

Table 7 shows statistical comparison obtained between modelled and BGC-Argo derived Kd(490), while **Figure 17** gives a comparison of vertical profiles. Compared to previous versions, bias is reduced while correlation is smaller.

It should be noted also that the surface estimates of Kd, obtained from BGC-ARGO, differ substantially from satellite-derived estimates, with biases between BGC-ARGO and satellite estimates larger than biases between BGC-ARGO and the model.

Table 7: *Model validation statistics obtained by comparing model and BG-ARGO predictions pairs for Kd (490). Validation metrics are defined in* ***Table 2****.*

	bias	perc. bias	RMSD	r_σ	Nash-Sutcliffe	Pearson
V202311	0.017 m^{-1}	16.22 %	0.08 m^{-1}	1.14	0.02	0.45
V202411	0.005 m^{-1}	5.82 %	0.11 m^{-1}	8.07	0.07	0.28
V202511	**0.004 m^{-1}**	**4.40 %**	**0.12 m^{-1}**	**6.62**	**0.11**	**0.27**

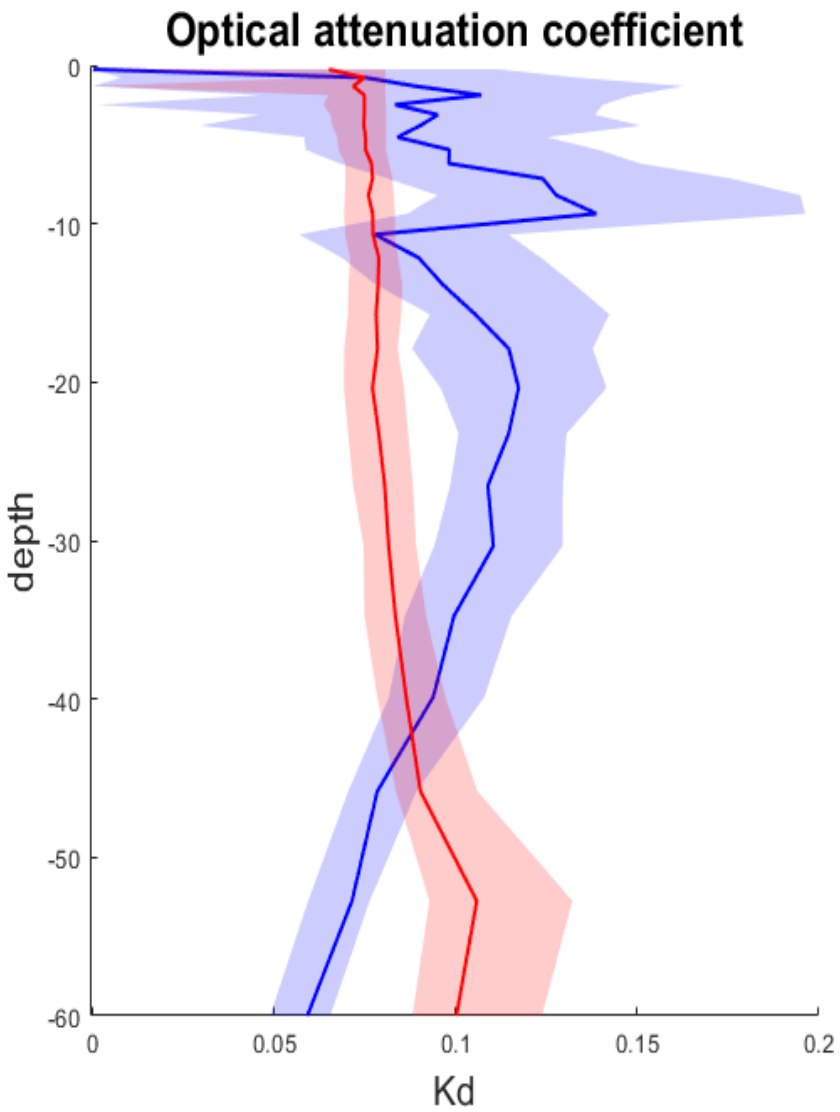

Figure 17: *Blue: BGC-Argo Kd_{490} estimated from BGC-ARGO irradiance. Red: Kd_{490} from model in version V202511. The solid line represents the median and shaded areas cover percentiles 25% to 75% (see Fig 1 for profile locations)*

IV.5.3 Reflectances

In this section, the model-derived reflectance is compared to the satellite-observed reflectance at 6 wavelenghts: 412, 443, 490, 510, 555 and 670 nm. Examples of region-averages are shown for the year 2024 in Fig 18, for regions 1 (open sea) and 5 (in front of the Danube delta).

Overall, the model reproduces reflectance values within the correct range. However, during summer, the coccolithophore bloom leads to increased reflectance observed by the satellite but not captured by the model as calcification is not explicitly represented.

At most wavelengths, a reflectance bias exists. It is generally positive, with model reflectance larger than the satellite one, and varies between 10 and 60%. In other regions, the bias is negative, such as in region 11 (Russian coastal area).

Furthermore, the satellite-observed reflectance presents more variability than the model estimates. The standard deviation ratio varies between 0.8 and 10, with the largest values at shorter wavelengths (412 and 443 nm) and in coastal regions. Part of this variability can be explained by the different number of pixels observed by the satellite in each region: cloud cover significantly reduces the number of valid satellite pixels, which can introduce more variability in the regional averages.

The uncertainty affecting the model reflectance is further detailed in Macé et al, 2024.

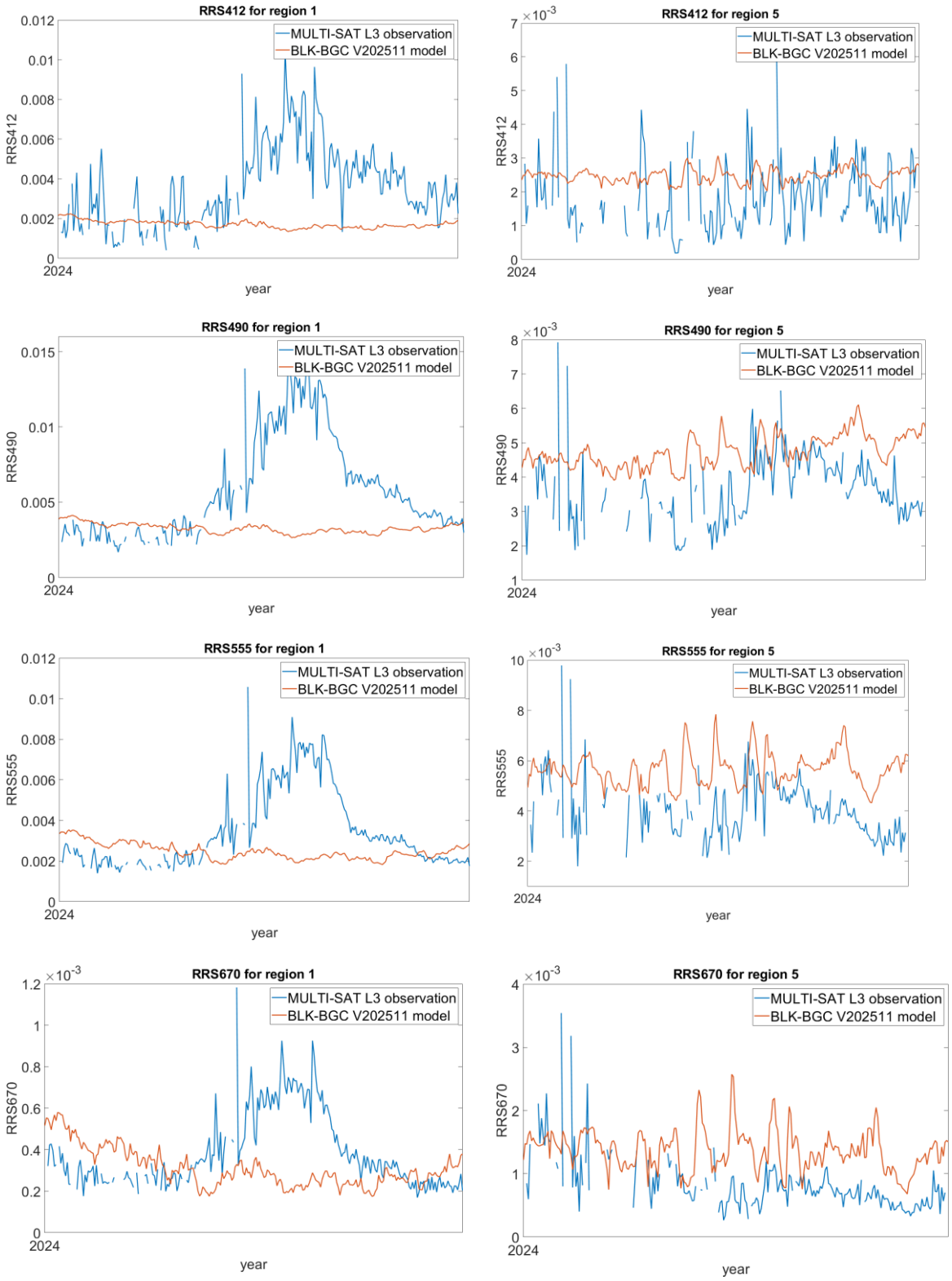

***Figure 18**. Comparison of model and satellite-observed reflectance at different wavelengths, for regions 1 and 5.*

V SYSTEM'S NOTICEABLE EVENTS, OUTGAES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. In includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality.

Up to version V202211, the hydrodynamic model underwent successive changes such as:

- Switching the physical forcing model to NEMO v3.6, then 4.0.6
- Change in the turbulence closure scheme and vertical diffusion coefficients.
- Improve the horizontal and vertical resolution
- Use of near-real time Danube freshwater discharge data

The BAMHBI biogeochemical model was also improved, e.g.:

- Addition of an alkalinity module.
- Addition of satellite chlorophyll observations to data assimilation.
- Use of time-varying atmospheric pCO2 concentrations, accounting for both interannual increases (as indicated by the IPCC) and seasonal variability.
- Addition of a new optics dataset containing 2 variables: photosynthetically active radiation (PAR) and optical attenuation coefficient (Kd).

Starting with version V202311, the following changes were implemented:

- Upgrade from Nemo 4.0.6 to Nemo 4.2
- New vertical discretisation, still using 59 z-layer levels, with higher resolution between 50-150 m depth and coarser below. The surface layer thickness remains 0.5m. The turbulence scheme has been changed from TKE to GLS
- The lateral diffusion is now computed using triads
- The surface heat exchange coefficients are now computed as a refined function of wind velocity
- Chlorophyll is now a state variable with one variable for each phytoplankton group
- Refinement in the algorithm of the sinking speed of particulate organic matter
- The data assimilation scheme is based on a SEEK filter using 3D EOFs to construct a reduced-rank model state error space. Similar to the common horizontal localization technique, aimed at suppressing spurious (horizontal) long-range correlations, the correction brought by the data assimilation procedure is also limited in the vertical. At V202311, this limitation is no longer static, it is now applied up to a depth itself computed as a function of the local MLD.

In V202311, the model configuration is exactly the same as for the corresponding multi-year product (007-005). The horizontal grid is also identical to the PHY and WAV products.

In V202411, the following changes were added:

- Addition of a radiative transfer model (RADTRANS)
- Assimilation of in situ dissolved oxygen observations

Furthermore, 2 new variables are added to the catalogue: zooplankton and oxygen-demand-units.

In V202511, the atmospheric deposition of nitrogen is obtained from the Copernicus service (CAMS) rather than from climatology. The nitrate deposition is approximately 3 times smaller in V202511 than in V202411, and the ammonium deposition is approximately 2.7 times smaller. Furthermore, the deposition is now variable in time and space; in particular, wet deposition events lead to concentrated depositions in areas and timespans corresponding to rainfall events.

Regarding rivers, the 11 major rivers are included in V202511 (six rivers in V202411). Except for the Danube, the river freshwater flux was obtained from climatology in V202411, whereas inV202511, it is obtained from EFAS version 5.

For the Danube River, the data supplier for the water flux remains unchanged: the National Hydrologic Institute of Romania (NIHWM), a partner in the BLK-MFC consortium.

For The Dnestr and Dnepr rivers, the fluxes from EFAS appear unrealistic during certain periods. Therefore, the temporal mean of the bias between EFAS and in situ observations, provided by the Ukrainian Academy of Sciences (UkrSc), was computed and subtracted from the EFAS data. It should be noted also that no significant bias exists between the NIHWM and UkrSc data for the Danube freshwater flux.

EFAS does not provide biogeochemical tracer concentration in the river. For the Danube, monthly concentration observations are obtained from the TransNational Monitoring Network (TNMN[13]). The average monthly concentrations are computed over the period 1996-2021 and extrapolated to present. The tracer discharge is then obtained by multiplying these concentrations by the real-time water flux.

For the other rivers, monthly climatological tracer discharges are obtained from the SESAME project (Ludwig et al, 2009).

These changes in data sources result in modifications to the water and nutrient discharge (mean values over 2022-2024) listed in Table 8. The values represent the ratio of the new value (V202511) to the old value (V202411).

Table 8. *Ratio of V202511 data over V202411 data.*

Rivers	Water flux	Nitrate	Ammonium	Dissolved nitrogen (labile compartment)	Phosphate	Silicate
Danube	1.00	0.63	0.57	1.53	3.46	1.46
Dnepr	0.40	0.49	0.44	0.32	0.56	0.39
Dnestr	1.05	2.02	1.82	2.74	1.52	1.37
Kizilirmak	0.38	0.14	0.13	0.23	0.07	0.21
Rioni	0.15	0.34	0.31	1.59	0.26	0.29
Sakarya	0.82	0.49	0.44	0.8	0.21	0.46

[13] TNMN - TransNational Monitoring Network | ICPDR - International Commission for the Protection of the Danube River

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

In V202205, the simulated oxycline was deeper than observed on ARGO data and the oxygen penetration depth was also too deep (**Table 3**). Since V202311, the oxycline depth in the model was much closer to ARGO observations, although a slight bias persists, with the model oxycline remaining too deep, particularly during summer. All EANs were largely improved compared to V202211 and previous versions. At V202411, thanks to the assimilation of in situ oxygen profiles, all EANs were improved further.

Regarding chlorophyll, compared to older model versions, V201907 model and satellite values were in much better agreement. In V202007 and V202211, assimilation of satellite chlorophyll observations brought the model even (much) closer to the observed values in the open sea.

In the V202311 model, chlorophyll representation improved further in the open ocean (both the surface value compared to satellites, and the vertical structure compared to BGC-ARGO). However, chlorophyll on the North-Western shelf is still over-estimated in summer.

In V202411, the biogeochemistry on the North-Western shelf improved considerably, as seen both in the EANs for chlorophyll and for optics. Although they underperform the model in the open sea, this underperformance is much less severe.

V202211 was the first model version with a sufficiently high vertical resolution to resolve the summer thermocline (around 10-15 m depth) and thus sustain a second, shallower DCM, distinct from the deeper DCM located around 50-70 m depth (just above the permanent halocline), in agreement with observations. This feature is conserved in V202311. V202211 also incorporated real-time Danube River discharge data to the model; the beneficial impact is preserved in V202311 and V202411. The addition of real-time water discharge for the other rivers, at V202511, results in very small changes in quality.

Since V202311, the depth of the deeper DCM is better represented thanks to refined vertical resolution in the corresponding depth range.

Nitrate concentrations at depth are better represented since version V202411.

Regarding optical variables, comparison of satellite and model time-series of the attenuation coefficient Kd(490nm) at the surface reveals an improvement since V202411, due to the inclusion of the spectral optical model. Comparison with in-situ measurements from BGC-ARGO floats reveals that Kd(490nm) has a lower bias, but also reduced correlation relative to V202311.

In V202511, the inclusion of near real-time freshwater fluxes for all rivers (in addition to the Danube) and of near real-time atmospheric deposition of nitrate and ammonium leads to a slight improvement for some variables and a marginal degradation for others.

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